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ZSDR0017 Program Specification

Order Item Report and Delivery Block for Large Orders

A draft spec written from the current code after taking over the program. What the program actually does differs from what its header says, so the current behavior is written down first, for the requester to confirm what should stay and what should change.

Item	Details
Document version	v0.1
Document status	Draft, awaiting confirmation from the requester
Request number	To be confirmed
Sources	ZSDR0017 source code, abap-archaeology program ID card

NOTE: *Sample program, written around common legacy ABAP problems, with no company data. English translation of the Traditional Chinese sample; abap-spec-writer itself writes specs in Traditional Chinese.*

1. Document Information

A draft based on the current code, awaiting the requester's confirmation

Item	Details
Program name	ZSDR0017
Request number	To be confirmed
Subject	List order items, with the option to set a delivery block on large orders
Document version	v0.1
Document status	Draft, awaiting confirmation from the requester
Developers	WLIN (created 2014), KCHEN (added the block feature 2016)
Requester	To be confirmed

This spec was written after taking over the program, and it is based on the current code.

What the program actually does differs from what its header says (see Section 2). So the current behavior is written down first, for the requester to confirm what should stay and what should change.

2. Background and Purpose

The header says "open orders", but the code filters no status at all

The two things the program does today:

1. It lists every item of standard orders in the Taiwan sales organization: order number, customer, material, quantity and amount.
2. When the user ticks "Apply block", any order worth more than 500,000 in total automatically gets a delivery block, so it cannot ship until someone releases it manually.

The problem the program was originally meant to solve (to be confirmed):

- The header says "Open Orders Report", and CR-0347 in 2015 asked to "show only items not yet shipped".
- But the current code has no filter on shipping status, so shipped and completed orders are listed too.
- The requester needs to define whether "open" means not shipped, not billed or not completed. That decides whether the report has to change.

The purpose of the block feature (to be confirmed): presumably, large orders need a credit or manager check before they ship. The code gives no source for the 500,000 threshold or the exempt customer.

Acceptance criteria (based on current behavior, to be revised once the requester confirms):

1. Given a creation date and customer range, the items in the report match the order items in the system.
2. With "Apply block" ticked, orders worth more than 500,000 in total show block code Z1 on the order.
3. With it unticked, the program only displays data and changes no orders.

3. Glossary

Seven terms to know before reading this document

Term	Meaning
Sales organization TW01	The organizational unit responsible for sales in Taiwan. The program only looks at this one.
Order types ZOR / ZOR2	Two company-defined order types. The program only looks at these two.
Delivery block	A flag on an order that "freezes" it: no delivery can be created until someone removes the flag manually.
Block code Z1	The delivery block reason code the program sets. Its business meaning is to be confirmed.
Order net value	The sum of all item amounts on one order, excluding tax, in the order's own currency.
BAPI	A standard interface provided by SAP. The program uses one to change orders, with the same effect as changing them manually on the order screen.
ALV	SAP's table-style report screen, which supports sorting, filtering and exporting to Excel.

4. Functional Scope

Three fields on the screen, and only "Apply block" changes data

4.1 Selection Screen

Field	Maps to	Required/Optional	Notes
Order creation date	Creation date on the order header	Optional	Accepts a range. Left empty, it reads all dates, which can be a lot of data, so always entering one is recommended.
Customer	Sold-to customer on the order header	Optional	Accepts multiple values or a range. Left empty, it means all customers.
Apply block	Checkbox	Optional	Orders are changed only when this is ticked; otherwise the program only shows the report. Its technical name on the screen is `p_x`, which says nothing about its purpose, so take extra care when running the program.

NOTE: *Sales organization TW01 and order types ZOR / ZOR2 are hardcoded and cannot be changed on the screen.*

4.2 Process Overview

1. Read order headers using the selection screen criteria plus the fixed sales organization and order types.
2. For each order, read the customer name and all items, and add one report line per item.
3. Add up the item amounts for the whole order at the same time.
4. If "Apply block" is ticked and the order meets the conditions, set delivery block Z1 and mark the order in the report.
5. Display the report.

5. Processing Logic

Blocked only when three conditions are all met, and failures get marked as blocked too

5.1 Reading Orders

The program only looks at orders in sales organization TW01 with order type ZOR or ZOR2, then applies the creation date and customer from the screen.

It does not filter on shipping, billing or completion status, and it does not exclude rejected items.

> **Core rule** **The report's scope is not "open orders"**
>
> Shipped orders, completed orders and orders with rejected items all appear in the report.
>
> Example: an order created in March and fully shipped in March still lists every item when you query March.
>
> Until the requester defines "open", do not use this report as an open-order list.

5.2 Building Report Lines

Each order item gets one line, showing order number, creation date, customer, customer name, item number, material, quantity and amount.

The customer name is read from the customer master.

Known limitation: if a customer cannot be read from the customer master, the line shows the customer name from the previous order instead of staying blank. This doesn't happen when the customer master is in good order, but if a name doesn't match the customer number, trust the customer number.

5.3 Delivery Block for Large Orders

Core rule **An order is blocked only when all three conditions are met**

1. "Apply block" is ticked on the selection screen.
2. The order net value is greater than 500,000 (exactly 500,000 is not blocked).
3. The customer is not 0000100023.

When all three are met, the program sets delivery block Z1 on the order and marks every line of that order as blocked.

Scenario	Order net value	Customer	"Apply block" ticked	Result
Standard large order	500,001	0000100001	Yes	Blocked
Exactly at the threshold	500,000	0000100001	Yes	Not blocked
Exempt customer	900,000	0000100023	Yes	Not blocked
Not ticked	900,000	0000100001	No	Not blocked, display only

Edge cases:

1. **Currency:** the threshold is compared in the order's own currency, with no conversion. An order of USD 20,000 is not blocked; an order of JPY 600,000 is. Whether the threshold should always be in TWD is to be confirmed.
2. **Re-run:** running the program again on a blocked order sets Z1 again, and the report marks it as blocked as usual. Nothing fails, but the report can't show which orders were newly blocked this time.
3. **Block fails:** if someone else has the order open for editing, or the block code can't be used, the block doesn't go through, yet the report still marks the order as blocked. For now, the only way to know whether it worked is to check the block code on the order.
4. **Interrupted run:** each order is saved right after it is blocked. If the program stops halfway, earlier orders are blocked and later ones aren't, and the report keeps no record of where it stopped.

6. Output Fields

One line per item, with no currency or unit column for amounts and quantities

Field	Data type	Notes
Order number	VBAK-VBELN	Each item of an order takes one line, so order numbers repeat.
Creation date	VBAK-ERDAT	The creation date on the order header, not the item's delivery date.
Customer	VBAK-KUNNR	Sold-to customer number. If it doesn't match the customer name, trust this field (see 5.2).
Customer name	KNA1-NAME1	The first name line in the customer master.
Item	VBAP-POSNR	Order item number.
Material	VBAP-MATNR	The material on this item.
Quantity	VBAP-KWMENG	Order quantity in the sales unit on the order. The report has no unit column.
Amount	VBAP-NETWR	Item net value, excluding tax, in the order currency. The report has no currency column, so amounts from orders in different currencies can't simply be added up.
Blocked	One character	X means the program submitted a block for this order during this run. It doesn't guarantee the block succeeded (see 5.3, edge case 3).

NOTE: *Whether the field structure ZSDS0017 used by the report screen matches the table above exactly is to be confirmed.*

7. Out of Scope

Ten items not implemented or still open, to be decided after the requester confirms

Item	Current state
Open-status filter	Not implemented. The definition of "open" is to be confirmed by the requester.
CR-0347, show only unshipped items	The condition is disabled in the code and never took effect.
Authorization check	None. Anyone who can run the program can block orders.
Handling of failed blocks	None. A failed block is still marked as blocked.
Currency conversion	None. The threshold is compared directly in the order currency.
Exempt customer 0000100023	Reason for the exemption is to be confirmed.
Source of the 500,000 threshold	To be confirmed, along with whether it still applies.
Only TW01 and ZOR / ZOR2	Reason to be confirmed.
Meaning of block code Z1	To be confirmed.
Background jobs	Whether a scheduled job runs this program, and whether it ticks "Apply block", is to be confirmed.